

Basics of building a data model in Power BI

Jason Romans

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Local User
Groups:



Jason Romans



Builder of Models
Senior BI Engineer

thedaxshepherd@gmail.com

<https://thedaxshepherd.net/>

<https://www.linkedin.com/in/jason-r-sql-jar>

- *Live in Nashville, TN*
- *Started as a SQL Server DBA*
- *Transitioned to the Microsoft BI Stack*
- *Work on everything from SQL Server Integration Services, SQL Server Database, Analysis Services, and Power BI*

Machine wash warm,
inside out, with like
colors. Only non-chlorine
bleach. Tumble dry low.
Medium iron.
Do not iron decoration.
Do not dry clean.



2EG

52% ALGODÓN

48% POLIÉSTER

ENSAMBLADO EN HONDURAS

DE COMPONENTES DE USA

2TG

52% COTON

48% POLYESTER

ASSEMBLÉ AU HONDURAS

DE COMPOSANTS USA

52% COTTON

48% POLYESTER

What is a model?

A Tabular model is a set of metadata like tables, relationships, measures, KPIs, calculation groups, hierarchies, translations, security roles, and many other elements that form the “semantic model” used to provide a navigation system in client tools like Power BI and Excel reports.

-- [SQLBI](#)

Daunting Task

- How do I bring in the data
- What should the model look like
- Get Data – Model Data – Visualize Data

What you
picture



Reality



Instructions

- Unfortunately, no step-by-step instructions
- Bulk Lego blocks
- Guidelines

Get Data

Where is the data



EXCEL SPREADSHEET

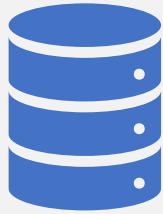


SQL SERVER

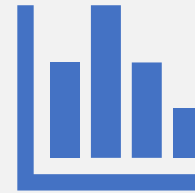


CSV

What form is the data in



SOURCE DATA THAT IS IN
SEPARATE TABLES

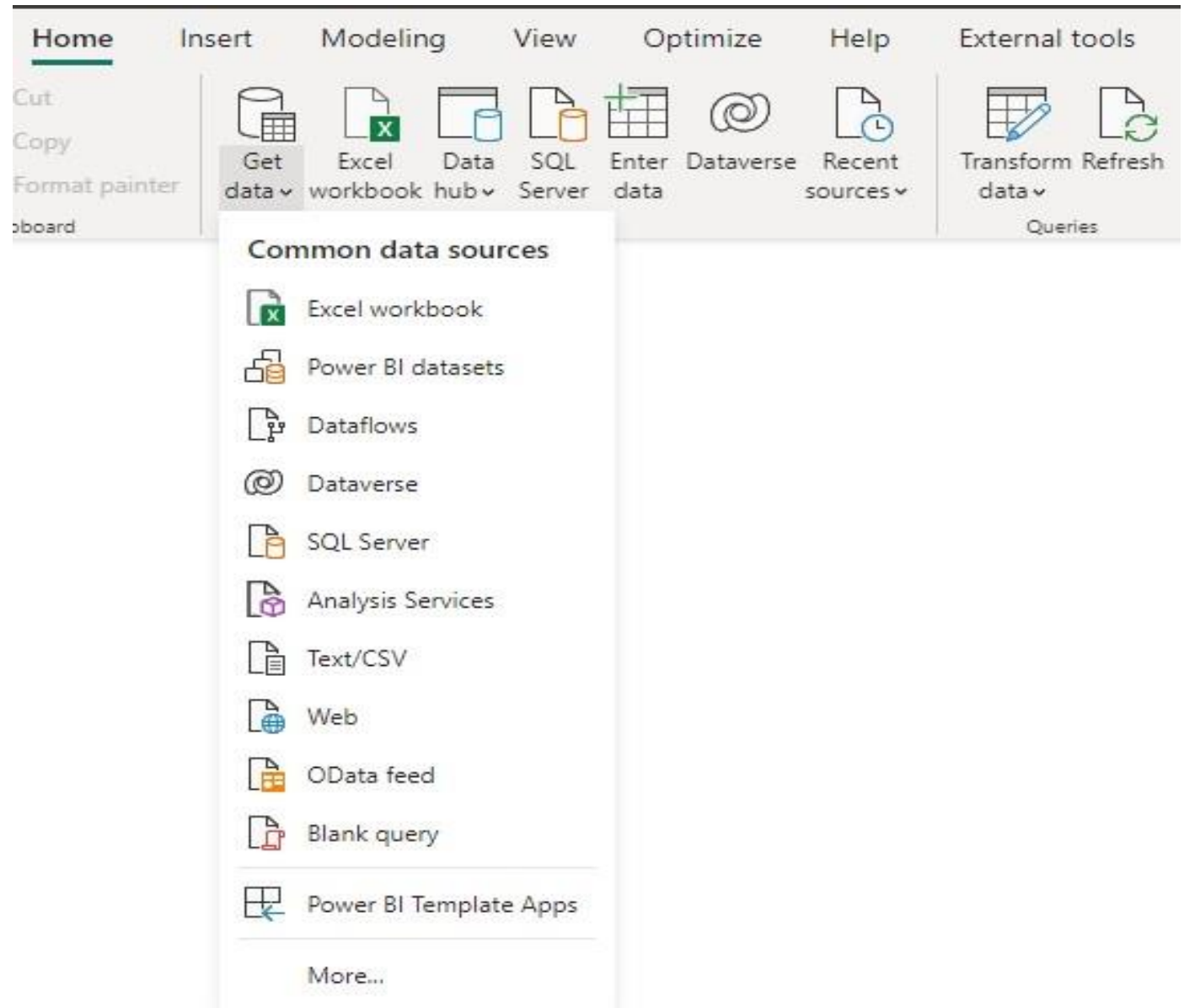


REPORT FORM WITH ALL
THE DATA COMBINED

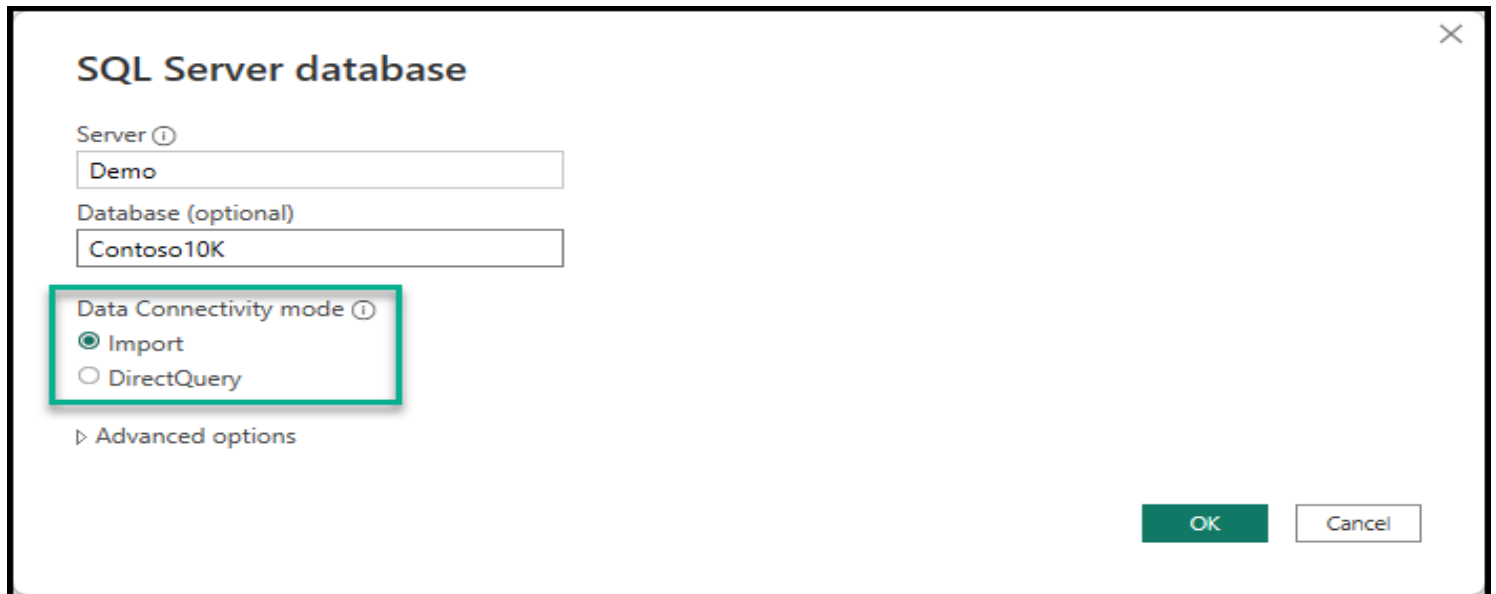
Toy Model Builder

- Start - picture of finished product
- Proceed through instructions step by step
- Unlike toy model – changes can be made
– no glue

Get Data



Import or DirectQuery



SQL Server database

Server ⓘ
Demo

Database (optional)
Contoso10K

Data Connectivity mode ⓘ

- ☒ Import
- ☐ DirectQuery

▸ Advanced options

OK Cancel

Select Tables or Views

Navigator

Display Options ▾

demo: Contoso10K [20]

☐

Currency Exchange

☒

Customer

☒

Date

☐

DIM_Customer

☐

DIM_Date

☐

DIM_Product

☐

DIM_Store

☐

FACT_Sales

☒

Product

☒

Sales

☒

Store

☐

Data.CurrencyExchange

☐

Data.Customer

☐

Data.Date

☐

Data.GeoLocations

☐

Data.OrderRows

☐

Data.Orders

☐

Data.Product

☐

Data.Store

☐

fn IsHolidav

Sales

Order Number	Line Number	Order Date	Delivery Date	CustomerKey	StoreKey	ProductKey
36600	0	1/1/2018	1/1/2018	1620763	540	
36700	0	1/2/2018	1/2/2018	1320351	480	
36700	1	1/2/2018	1/2/2018	1320351	480	
36700	2	1/2/2018	1/2/2018	1320351	480	
36700	3	1/2/2018	1/2/2018	1320351	480	
36700	4	1/2/2018	1/2/2018	1320351	480	
36701	0	1/2/2018	1/2/2018	2077250	585	
36800	0	1/3/2018	1/10/2018	1743422	999999	
36800	1	1/3/2018	1/10/2018	1743422	999999	
36800	2	1/3/2018	1/10/2018	1743422	999999	
36800	3	1/3/2018	1/10/2018	1743422	999999	
36801	0	1/3/2018	1/3/2018	1013948	390	
36801	1	1/3/2018	1/3/2018	1013948	390	
36801	2	1/3/2018	1/3/2018	1013948	390	
36801	3	1/3/2018	1/3/2018	1013948	390	
36900	0	1/4/2018	1/4/2018	544050	220	
36901	0	1/4/2018	1/14/2018	485447	999999	
36902	0	1/4/2018	1/17/2018	1075463	999999	
37000	0	1/5/2018	1/5/2018	1319917	550	
37000	1	1/5/2018	1/5/2018	1319917	550	
37000	2	1/5/2018	1/5/2018	1319917	550	
37000	3	1/5/2018	1/5/2018	1319917	550	
37100	0	1/6/2018	1/6/2018	1107123	380	
37100	1	1/6/2018	1/6/2018	1107123	380	

Select Related Tables

Load

Transform Data

Cancel

Load or Transform Data

Navigator

Display Options ▾

demo: Contoso10K [20]

☐ Currency Exchange

☒ Customer

☒ Date

☐ DIM_Customer

☐ DIM_Date

☐ DIM_Product

☐ DIM_Store

☐ FACT_Sales

☒ Product

☒ Sales

☒ Store

☐ Data.CurrencyExchange

☐ Data.Customer

☐ Data.Date

☐ Data.GeoLocations

☐ Data.OrderRows

☐ Data.Orders

☐ Data.Product

☐ Data.Store

☐ fn IsHoliday

Sales

Order Number	Line Number	Order Date	Delivery Date	CustomerKey	StoreKey	ProductKey
36600	0	1/1/2018	1/1/2018	1620763	540	
36700	0	1/2/2018	1/2/2018	1320351	480	
36700	1	1/2/2018	1/2/2018	1320351	480	
36700	2	1/2/2018	1/2/2018	1320351	480	
36700	3	1/2/2018	1/2/2018	1320351	480	
36700	4	1/2/2018	1/2/2018	1320351	480	
36701	0	1/2/2018	1/2/2018	2077250	585	
36800	0	1/3/2018	1/10/2018	1743422	999999	
36800	1	1/3/2018	1/10/2018	1743422	999999	
36800	2	1/3/2018	1/10/2018	1743422	999999	
36800	3	1/3/2018	1/10/2018	1743422	999999	
36801	0	1/3/2018	1/3/2018	1013948	390	
36801	1	1/3/2018	1/3/2018	1013948	390	
36801	2	1/3/2018	1/3/2018	1013948	390	
36801	3	1/3/2018	1/3/2018	1013948	390	
36900	0	1/4/2018	1/4/2018	544050	220	
36901	0	1/4/2018	1/14/2018	485447	999999	
36902	0	1/4/2018	1/17/2018	1075463	999999	
37000	0	1/5/2018	1/5/2018	1319917	550	
37000	1	1/5/2018	1/5/2018	1319917	550	
37000	2	1/5/2018	1/5/2018	1319917	550	
37000	3	1/5/2018	1/5/2018	1319917	550	
37100	0	1/6/2018	1/6/2018	1107123	380	
37100	1	1/6/2018	1/6/2018	1107123	380	

Select Related Tables

Load

Transform Data

Cancel

Power Query Editor

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply Close New Source Recent Sources Enter Data Data source settings Data Sources Manage Parameters Parameters Refresh Preview Advanced Editor Manage Query Choose Columns Remove Columns Manage Columns Keep Rows Remove Rows Reduce Rows Sort Split Column Group By Data Type: Whole Number Use First Row as Headers Replace Values Combine Text Analytics Vision Azure Machine Learning AI Insights

Queries [4]

- Store
- Product
- Sales
- Customer

fx = Source[[Schema="dbo",Item="Sales"]][Data]

	Order Number	Line Number	Order Date	Delivery Date	CustomerKey
1	36600	0	1/1/2018	1/1/2018	
2	36700	0	1/2/2018	1/2/2018	
3	36700	1	1/2/2018	1/2/2018	
4	36700	2	1/2/2018	1/2/2018	
5	36700	3	1/2/2018	1/2/2018	
6	36700	4	1/2/2018	1/2/2018	
7	36701	0	1/2/2018	1/2/2018	
8	36800	0	1/3/2018	1/10/2018	
9	36800	1	1/3/2018	1/10/2018	
10	36800	2	1/3/2018	1/10/2018	
11	36800	3	1/3/2018	1/10/2018	
12	36801	0	1/3/2018	1/3/2018	
13	36801	1	1/3/2018	1/3/2018	
14	36801	2	1/3/2018	1/3/2018	
15	36801	3	1/3/2018	1/3/2018	
16	36900	0	1/4/2018	1/4/2018	
17	36901	0	1/4/2018	1/14/2018	
18	36902	0	1/4/2018	1/17/2018	
19	37000	0	1/5/2018	1/5/2018	
20	37000	1	1/5/2018	1/5/2018	
21	37000	2	1/5/2018	1/5/2018	
22	37000	3	1/5/2018	1/5/2018	
23	37100	0	1/6/2018	1/6/2018	
24	37100	1	1/6/2018	1/6/2018	
25					

Query Settings

PROPERTIES

Name

Sales

All Properties

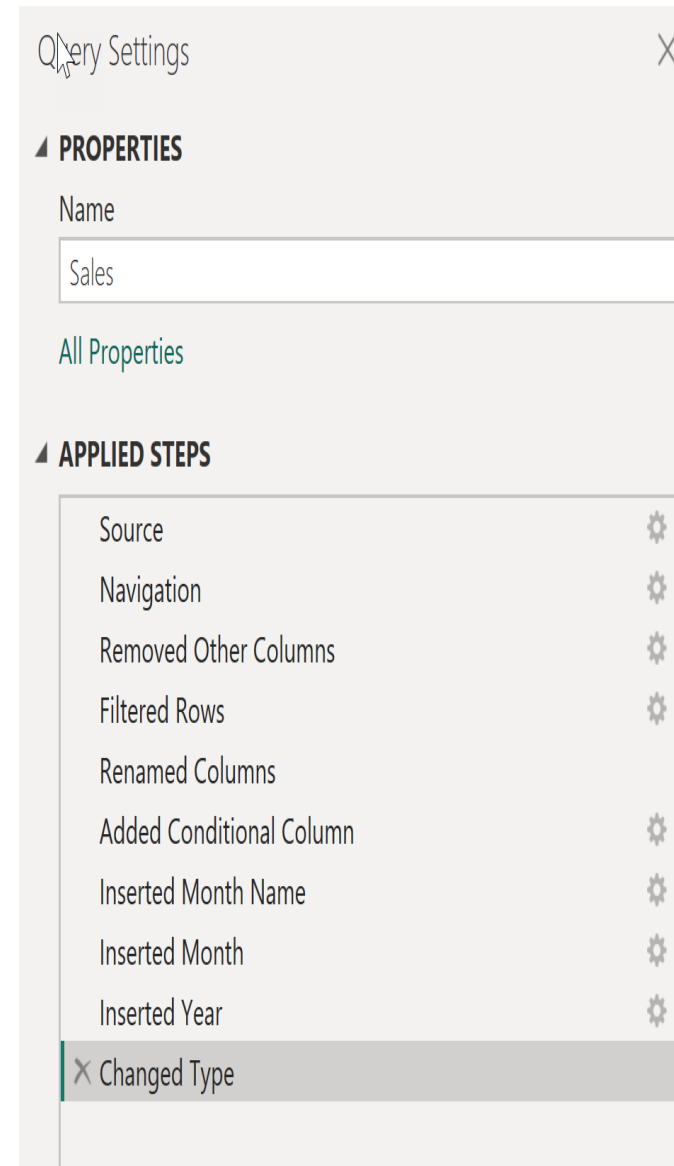
APPLIED STEPS

- Source
- Navigation

13 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 2:04 PM

Power Query Steps like a Recipe



Promote First Row to Headers (CSV)

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By

Use First Row as Headers

Transpose

Reverse Rows

Count Rows

Data Type: Text

Detect Data Type

Rename

Replace Values

Fill

Pivot Column

Unpivot Columns

Move

Convert to List

Any Column

Use First Row as Headers

Use Headers as First Row

Promote the first row of this table into column headers.

Sales

	Column1	Column2
1	Order Number	Line Number
2	36600	0
3	36700	0
4	36700	1
5	36700	2

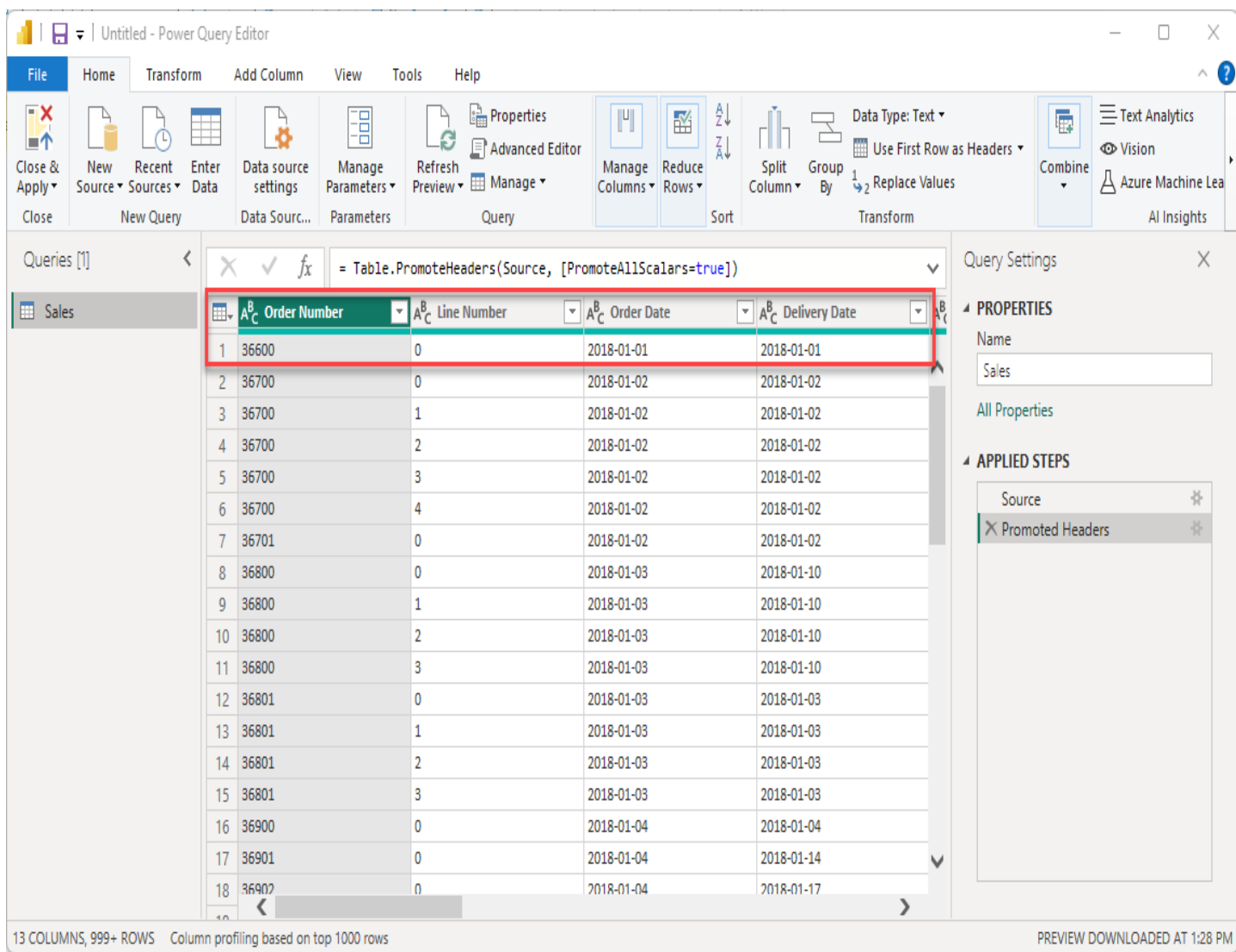
Promote First Row to Headers (CSV)

The screenshot shows the Power Query Editor interface. The ribbon includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The 'Transform' tab is active, showing options like 'Use First Row as Headers' and 'Replace Values'. The 'Queries' pane on the left shows a query named 'Sales'. The main area displays a table with 13 columns and 999+ rows. The first row is highlighted in green, indicating it has been promoted to headers. The formula bar shows the query definition: `= Csv.Document(File.Contents("C:\ContosoDataFiles\Sales.csv"),[Delimiter=",",`. The 'Query Settings' pane on the right shows the 'Name' property set to 'Sales' and the 'Applied Steps' list containing 'Source'.

Order Number	Line Number	Order Date	Delivery Date
36600	0	2018-01-01	2018-01-01
36700	0	2018-01-02	2018-01-02
36700	1	2018-01-02	2018-01-02
36700	2	2018-01-02	2018-01-02
36700	3	2018-01-02	2018-01-02
36700	4	2018-01-02	2018-01-02
36701	0	2018-01-02	2018-01-02
36800	0	2018-01-03	2018-01-10
36800	1	2018-01-03	2018-01-10
36800	2	2018-01-03	2018-01-10
36800	3	2018-01-03	2018-01-10
36801	0	2018-01-03	2018-01-03
36801	1	2018-01-03	2018-01-03
36801	2	2018-01-03	2018-01-03
36801	3	2018-01-03	2018-01-03
36900	0	2018-01-04	2018-01-04
36901	0	2018-01-04	2018-01-14

13 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 1:27 PM

Promote First Row to Headers (CSV)



The screenshot displays the Power Query Editor interface. The formula bar at the top shows the query definition: `= Table.PromoteHeaders(Source, [PromoteAllScalars=true])`. The data table below has four columns: Order Number, Line Number, Order Date, and Delivery Date. The first row of data is highlighted with a red border, indicating it has been promoted to the header row. The 'APPLIED STEPS' pane on the right shows the 'Promoted Headers' step.

	Order Number	Line Number	Order Date	Delivery Date
1	36600	0	2018-01-01	2018-01-01
2	36700	0	2018-01-02	2018-01-02
3	36700	1	2018-01-02	2018-01-02
4	36700	2	2018-01-02	2018-01-02
5	36700	3	2018-01-02	2018-01-02
6	36700	4	2018-01-02	2018-01-02
7	36701	0	2018-01-02	2018-01-02
8	36800	0	2018-01-03	2018-01-10
9	36800	1	2018-01-03	2018-01-10
10	36800	2	2018-01-03	2018-01-10
11	36800	3	2018-01-03	2018-01-10
12	36801	0	2018-01-03	2018-01-03
13	36801	1	2018-01-03	2018-01-03
14	36801	2	2018-01-03	2018-01-03
15	36801	3	2018-01-03	2018-01-03
16	36900	0	2018-01-04	2018-01-04
17	36901	0	2018-01-04	2018-01-14
18	36902	0	2018-01-04	2018-01-17

13 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 1:28 PM

Remove Other Columns

The screenshot displays the Microsoft Power Query Editor window titled "Untitled - Power Query Editor". The ribbon includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The "Remove Columns" button in the Transform tab is active, and its dropdown menu is open, showing options: "Remove Columns", "Remove Other Columns", "Keep Rows", and "Remove Rows". The main data area shows a table with 25 rows and 5 columns: Order Number, Line Number, Order Date, Delivery Date, and CustomerKey. The right sidebar contains the "Query Settings" pane, which is divided into "PROPERTIES" (Name: Sales) and "APPLIED STEPS" (Source, Navigation).

	Order Number	Line Number	Order Date	Delivery Date	CustomerKey
1	36600	0	1/1/2018	1/1/2018	1620763
2	36700	0	1/2/2018	1/2/2018	1320351
3	36700	1	1/2/2018	1/2/2018	1320351
4	36700	2	1/2/2018	1/2/2018	1320351
5	36700	3	1/2/2018	1/2/2018	1320351
6	36700	4	1/2/2018	1/2/2018	1320351
7	36701	0	1/2/2018	1/2/2018	2077250
8	36800	0	1/3/2018	1/10/2018	1743422
9	36800	1	1/3/2018	1/10/2018	1743422
10	36800	2	1/3/2018	1/10/2018	1743422
11	36800	3	1/3/2018	1/10/2018	1743422
12	36801	0	1/3/2018	1/3/2018	1013948
13	36801	1	1/3/2018	1/3/2018	1013948
14	36801	2	1/3/2018	1/3/2018	1013948
15	36801	3	1/3/2018	1/3/2018	1013948
16	36900	0	1/4/2018	1/4/2018	544050
17	36901	0	1/4/2018	1/14/2018	485447
18	36902	0	1/4/2018	1/17/2018	1075463
19	37000	0	1/5/2018	1/5/2018	1319917
20	37000	1	1/5/2018	1/5/2018	1319917
21	37000	2	1/5/2018	1/5/2018	1319917
22	37000	3	1/5/2018	1/5/2018	1319917
23	37100	0	1/6/2018	1/6/2018	1107123
24	37100	1	1/6/2018	1/6/2018	1107123
25					

13 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED ON TUESDAY

Filter Rows

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply Close New Source Recent Sources Enter Data Data source settings Data Sources Manage Parameters Parameters Refresh Preview Advanced Editor Query Choose Columns Remove Columns Manage Columns Keep Rows Remove Rows Reduce Rows Sort Split Column Group By Use First Row as Headers Replace Values Transform Data Type: Date

Combine Text Analytics Vision Azure Machine Learning All Insights

Queries [1] Sales

Order Number Line Number Order Date Delivery Date CustomerKey

1 36600 0 1/1/2018 1/1/2018 1620763

2 36700 0 1/2/2018 1/2/2018 1320351

3 36700 1 1/2/2018 1/2/2018 1320351

4 36700 2 1/2/2018 1/2/2018 1320351

5 1320351

6 1320351

7 2077250

8 1743422

9 1743422

10 1743422

11 1743422

12 1013948

13 1013948

14 1013948

15 1013948

16 544050

17 485447

18 1075463

19 1319917

20 37000 1 1/5/2018 1/5/2018 1319917

21 37000 2 1/5/2018 1/5/2018 1319917

22 37000 3 1/5/2018 1/5/2018 1319917

23 37100 0 1/6/2018 1/6/2018 1107123

24 37100 1 1/6/2018 1/6/2018 1107123

25

Filter Rows

Apply one or more filter conditions to the rows in this table.

☒ Basic ☐ Advanced

Keep rows where 'Order Date'

is after 6/13/2018

☒ And ☐ Or

Enter or select a value

OK Cancel

13 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED ON TUESDAY

Set Data Types

- Correct types matter
 - Same type for relationships
- Performance
- Calculations

ProductKey as a String

Name	Cardinality	Total Size ↓	Data	Dictionary	Hier Size	Encoding	Data Type	RI Violations	User Hier Size	Rel Size	% Table
Product	2,517	15,142,472	27,520	15,043,768	71,184	Many	-	-	0	0	
Product Name	2,517	1,128,480	4,160	1,104,176	20,144	HASH	String	-	-	-	7.45%
Product Code	2,517	1,127,544	4,160	1,103,240	20,144	HASH	String	-	-	-	7.45%
ProductKey	2,517	1,127,416	4,160	1,103,112	20,144	HASH	String	-	-	-	7.45%
Unit Cost	480	1,076,408	3,008	1,069,544	3,856	HASH	String	-	-	-	7.11%
Unit Price	426	1,075,520	3,008	1,069,088	3,424	HASH	String	-	-	-	7.10%
Weight	297	1,073,176	2,736	1,068,056	2,384	HASH	String	-	-	-	7.09%
Subcategory	32	1,067,560	1,352	1,065,936	272	HASH	String	-	-	-	7.05%
Subcategory Code	32	1,067,560	1,352	1,065,936	272	HASH	String	-	-	-	7.05%
Color	16	1,067,144	1,192	1,065,808	144	HASH	String	-	-	-	7.05%
Brand	11	1,066,464	600	1,065,768	96	HASH	String	-	-	-	7.04%
Manufacturer	11	1,066,464	600	1,065,768	96	HASH	String	-	-	-	7.04%
Category	8	1,066,288	464	1,065,744	80	HASH	String	-	-	-	7.04%
Category Code	8	1,066,288	464	1,065,744	80	HASH	String	-	-	-	7.04%
Weight Unit Measure	4	1,065,896	136	1,065,712	48	HASH	String	-	-	-	7.04%

ProductKey as a Whole Number

Name	Cardinality	Total Size ↓	Data	Dictionary	Hier Size	Encoding	Data Type	RI Violations	User Hier Size	Rel Size	% Table
Product	2,517	14,115,268	27,520	14,016,564	71,184	Many	-	-	0	0	
Product Name	2,517	1,128,480	4,160	1,104,176	20,144	HASH	String	-	-	-	7.99%
Product Code	2,517	1,127,544	4,160	1,103,240	20,144	HASH	String	-	-	-	7.99%
Unit Cost	480	1,076,408	3,008	1,069,544	3,856	HASH	String	-	-	-	7.63%
Unit Price	426	1,075,520	3,008	1,069,088	3,424	HASH	String	-	-	-	7.62%
Weight	297	1,073,176	2,736	1,068,056	2,384	HASH	String	-	-	-	7.60%
Subcategory	32	1,067,560	1,352	1,065,936	272	HASH	String	-	-	-	7.56%
Subcategory Code	32	1,067,560	1,352	1,065,936	272	HASH	String	-	-	-	7.56%
Color	16	1,067,144	1,192	1,065,808	144	HASH	String	-	-	-	7.56%
Brand	11	1,066,464	600	1,065,768	96	HASH	String	-	-	-	7.56%
Manufacturer	11	1,066,464	600	1,065,768	96	HASH	String	-	-	-	7.56%
Category	8	1,066,288	464	1,065,744	80	HASH	String	-	-	-	7.55%
Category Code	8	1,066,288	464	1,065,744	80	HASH	String	-	-	-	7.55%
Weight Unit Measure	4	1,065,896	136	1,065,712	48	HASH	String	-	-	-	7.55%
ProductKey	2,517	100,212	4,160	75,908	20,144	HASH	Int64	-	-	-	0.71%

Cardinality

Name	Cardinality	Total Size ↓	Data	Dictionary	Hier Size	Encoding	Data Type	RI Violations	User Hier Size	Rel Size	% Table
Product	2,517	15,142,472	27,520	15,043,768	71,184	Many	-	-	0	0	
Product Name	2,517	1,128,480	4,160	1,104,176	20,144	HASH	String	-	-	-	7.45%
Product Code	2,517	1,127,544	4,160	1,103,240	20,144	HASH	String	-	-	-	7.45%
ProductKey	2,517	1,127,416	4,160	1,103,112	20,144	HASH	String	-	-	-	7.45%
Unit Cost	480	1,076,408	3,008	1,069,544	3,856	HASH	String	-	-	-	7.11%
Unit Price	426	1,075,520	3,008	1,069,088	3,424	HASH	String	-	-	-	7.10%
Weight	297	1,073,176	2,736	1,068,056	2,384	HASH	String	-	-	-	7.09%
Subcategory	32	1,067,560	1,352	1,065,936	272	HASH	String	-	-	-	7.05%
Subcategory Code	32	1,067,560	1,352	1,065,936	272	HASH	String	-	-	-	7.05%
Color	16	1,067,144	1,192	1,065,808	144	HASH	String	-	-	-	7.05%
Brand	11	1,066,464	600	1,065,768	96	HASH	String	-	-	-	7.04%
Manufacturer	11	1,066,464	600	1,065,768	96	HASH	String	-	-	-	7.04%
Category	8	1,066,288	464	1,065,744	80	HASH	String	-	-	-	7.04%
Category Code	8	1,066,288	464	1,065,744	80	HASH	String	-	-	-	7.04%
Weight Unit Measure	4	1,065,896	136	1,065,712	48	HASH	String	-	-	-	7.04%

Cardinality – Unique Values

- The less unique values the better
- Do you really need Date and Time
- Does the time need to include seconds
 - What about morning, afternoon, night

Custom Columns

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Store Size

	Column Name	Operator	Value ⓘ			Output ⓘ		
If	Square Meters ▾	equals ▾	ABC 123 ▾	null	Then	ABC 123 ▾	N/A	...
Else If	Square Meters ▾	is less than or equ... ▾	ABC 123 ▾	500	Then	ABC 123 ▾	Small	...
Else If	Square Meters ▾	is less than or equ... ▾	ABC 123 ▾	1000	Then	ABC 123 ▾	Medium	

Add Clause

Else ⓘ

ABC 123 ▾

Large

OK

Cancel

Add Column from Example

Add Column From Examples
Enter sample values to create a new column (Ctrl+Enter to apply).

OK Cancel

	☒ A ^B _C Zip Code	☐ A ^B _C Country Code	☐ A ^B _C Country	☐ A ^B _C Continent	☒ A ^B _C Birthday	
1	13901	US	United States	North America	5/14/1982 0:00	
2	99362	US	United States	North America	6/25/1947 0:00	
3	47274	US	United States	North America	8/29/1942 0:00	
4	48607	US	United States	North America	8/16/1946 0:00	
5	36330	US	United States	North America	12/19/1992 0:00	
6	76102	US	United States	North America	3/3/1946 0:00	
7	91762	US	United States	North America	10/3/2001 0:00	
8	20036	US	United States	North America	3/15/1976 0:00	
9	07501	US	United States	North America	4/9/1979 0:00	
10	52501	US	United States	North America	9/3/1947 0:00	
11	78040	US	United States	North America	10/3/1995 0:00	
12	71201	US	United States	North America	2/13/1997 0:00	
13	53916	US	United States	North America	4/28/1964 0:00	
14	85283	US	United States	North America	1/2/1953 0:00	
15	63011	US	United States	North America	1/13/1956 0:00	
16	97205	US	United States	North America	12/21/1971 0:00	
17	39211	US	United States	North America	11/24/1998 0:00	
18	85364	US	United States	North America	10/12/1946 0:00	
19	03276	US	United States	North America	5/29/1938 0:00	
20	95204	US	United States	North America	2/13/1982 0:00	
21	85724	US	United States	North America	11/19/1943 0:00	
22	97205	US	United States	North America	10/19/1975 0:00	

Column1

- 5/14/1982 0:00 (Birthday)
- 14 (Day from Birthday)
- 5 (Day of Week from Birthday)
- Friday (Day of Week Name from Birthday)
- 134 (Day of Year from Birthday)
- 31 (Days in Month from Birthday)
- 1982-05-14T23:59:59.9999999 (End of Day from Birthday)
- 1982-05-31T23:59:59.9999999 (End of Month from Birthday)
- 1982-06-30T23:59:59.9999999 (End of Quarter from Birthday)
- 1982-05-15T23:59:59.9999999 (End of Week from Birthday)
- 1982-12-31T23:59:59.9999999 (End of Year from Birthday)
- 5 (Month from Birthday)
- May (Month Name from Birthday)
- 2 (Quarter of Year from Birthday)
- 5/1/1982 12:00:00 AM (Start of Month from Birthday)
- 4/1/1982 12:00:00 AM (Start of Quarter from Birthday)
- 5/9/1982 12:00:00 AM (Start of Week from Birthday)
- 1/1/1982 12:00:00 AM (Start of Year from Birthday)
- 3 (Week of Month from Birthday)
- 20 (Week of Year from Birthday)

Query Settings

PROPERTIES

Name

Customer

All Properties

Add Column from Example

Add Column From Examples

Enter sample values to create a new column (Ctrl+Enter to apply).

Transform: `Date.Year(DateTime.From([Birthday]))`

OK

Cancel

	☒ Zip Code	☐ Country Code	☐ Country	☐ Continent	☒ Birthday	Year
1	13901	US	United States	North America	5/14/1982 0:00	1982
2	99362	US	United States	North America	6/25/1947 0:00	1947
3	47274	US	United States	North America	8/29/1942 0:00	1942
4	48607	US	United States	North America	8/16/1946 0:00	1946
5	36330	US	United States	North America	12/19/1992 0:00	1992
6	76102	US	United States	North America	3/3/1946 0:00	1946
7	91762	US	United States	North America	10/3/2001 0:00	2001
8	20036	US	United States	North America	3/15/1976 0:00	1976
9	07501	US	United States	North America	4/9/1979 0:00	1979
10	52501	US	United States	North America	9/3/1947 0:00	1947
11	78040	US	United States	North America	10/3/1995 0:00	1995
12	71201	US	United States	North America	2/13/1997 0:00	1997
13	53916	US	United States	North America	4/28/1964 0:00	1964
14	85283	US	United States	North America	1/2/1953 0:00	1953
15	63011	US	United States	North America	1/13/1956 0:00	1956
16	97205	US	United States	North America	12/21/1971 0:00	1971
17	39211	US	United States	North America	11/24/1998 0:00	1998
18	85364	US	United States	North America	10/12/1946 0:00	1946
19	03276	US	United States	North America	5/29/1938 0:00	1938
20	95204	US	United States	North America	2/13/1982 0:00	1982
21	85724	US	United States	North America	11/19/1943 0:00	1943
22	97205	US	United States	North America	10/19/1975 0:00	1975
23	48066	US	United States	North America	8/13/1995 0:00	1995
24	38049	US	United States	North America	11/13/1939 0:00	1939
25	13202	US	United States	North America	11/24/1961 0:00	1961
26	49546	US	United States	North America	8/24/1940 0:00	1940
27	60014	US	United States	North America	6/17/1985 0:00	1985
28	32501	US	United States	North America	4/16/1951 0:00	1951
29	99336	US	United States	North America	6/22/1999 0:00	1999

Get Data Demo

Query Folding

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Replace Values Data Type: Any Use First Row as Headers Combine Text Analytics Vision Azure Machine Learning AI Insights

Queries [1] Sales

Table.SelectRows(#"Removed Other Columns", each [Order Date] > #date(2018, 6, 13))

	Order Number	Line Number	Order Date	Delivery Date	CustomerKey
1	53000	0	6/14/2018	6/14/2018	1365902
2	53000	1	6/14/2018	6/14/2018	1365902
3	53200	0	6/16/2018	6/16/2018	1656264
4	53200	1	6/16/2018	6/16/2018	1656264
5	53200	2	6/16/2018	6/16/2018	1656264
6	53200	3	6/16/2018	6/16/2018	1656264
7	53600	0	6/20/2018	7/2/2018	1799984
8	53700	0	6/21/2018	6/21/2018	1373034
9	53700	1	6/21/2018	6/21/2018	1373034
10	53700	2	6/21/2018	6/21/2018	1373034
11	53900	0	6/23/2018	6/23/2018	326607
12	53900	1	6/23/2018	6/23/2018	326607
13	53900	2	6/23/2018	6/23/2018	326607
14	54300	0	6/27/2018	6/27/2018	687715
15	54400	0	6/28/2018	6/28/2018	1734949
16	54400	1	6/28/2018	6/28/2018	1734949
17	54600	0	6/30/2018	6/30/2018	1043780
18	54600	1	6/30/2018	6/30/2018	1043780
19	55000	0	7/4/2018	7/4/2018	98175
20	55000	1	7/4/2018	7/4/2018	98175
21	55000	2	7/4/2018	7/4/2018	98175
22	55000	3	7/4/2018	7/4/2018	98175
23	55000	4	7/4/2018	7/4/2018	98175
24	55000	5	7/4/2018	7/4/2018	98175
25	55000	6	7/4/2018	7/4/2018	98175

5 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 2:06 PM

Query Settings

PROPERTIES

Name: Sales

APPLIED STEPS

Source

Navigation

Removed Other Columns

Filter

Edit Settings

Rename

Delete

Delete Until End

Insert Step After

Move before

Move after

Extract Previous

View Native Query

Diagnose

Properties...

Native Query

Native Query

```
select [_].[Order Number],  
       [_].[Line Number],  
       [_].[Order Date],  
       [_].[Delivery Date],  
       [_].[CustomerKey]  
from  
(  
    select [Order Number],  
           [Line Number],  
           [Order Date],  
           [Delivery Date],  
           [CustomerKey]  
    from [dbo].[Sales] as [$Table]  
) as [_]  
where [_].[Order Date] > convert(date, '2018-06-13')
```

OK

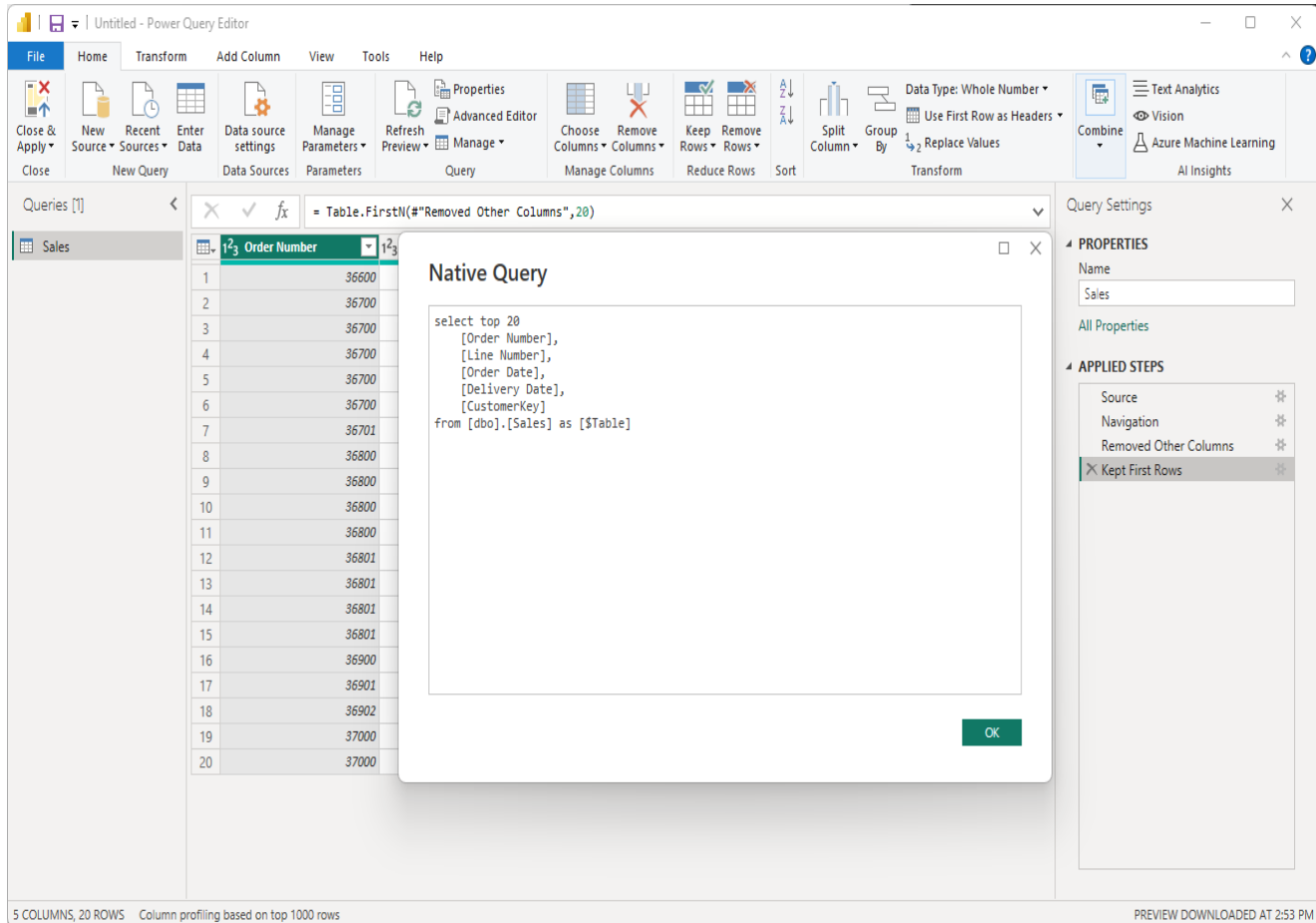
Breaking (Bad) Query Folding

The screenshot shows the Power Query Editor interface. The main area displays a table with 20 rows and 5 columns: Order Number, Line Number, Order Date, Delivery Date, and CustomerKey. The formula bar at the top shows the query definition: `= Table.LastN(#"Removed Other Columns", 20)`. The 'APPLIED STEPS' pane on the right shows the step 'Removed Other Columns' selected, with a context menu open showing options like 'View Native Query'.

	Order Number	Line Number	Order Date	Delivery Date	CustomerKey
1	115302	1	2/27/2020	2/27/2020	924770
2	115400	0	2/28/2020	2/28/2020	353571
3	115400	1	2/28/2020	2/28/2020	353571
4	115401	0	2/28/2020	3/8/2020	1358107
5	115500	0	2/29/2020	2/29/2020	345884
6	115501	0	2/29/2020	3/2/2020	1485625
7	115501	1	2/29/2020	3/2/2020	1485625
8	115501	2	2/29/2020	3/2/2020	1485625
9	115501	3	2/29/2020	3/2/2020	1485625
10	115501	4	2/29/2020	3/2/2020	1485625
11	115501	5	2/29/2020	3/2/2020	1485625
12	115501	6	2/29/2020	3/2/2020	1485625
13	115502	0	2/29/2020	2/29/2020	982512
14	115502	1	2/29/2020	2/29/2020	982512
15	115502	2	2/29/2020	2/29/2020	982512
16	115502	3	2/29/2020	2/29/2020	982512
17	115502	4	2/29/2020	2/29/2020	982512
18	115502	5	2/29/2020	2/29/2020	982512
19	115502	6	2/29/2020	2/29/2020	982512
20	115800	0	3/3/2020	3/3/2020	237000

Keep bottom 20 rows breaks query folding
SQL Server does not have a bottom clause

Query Folding

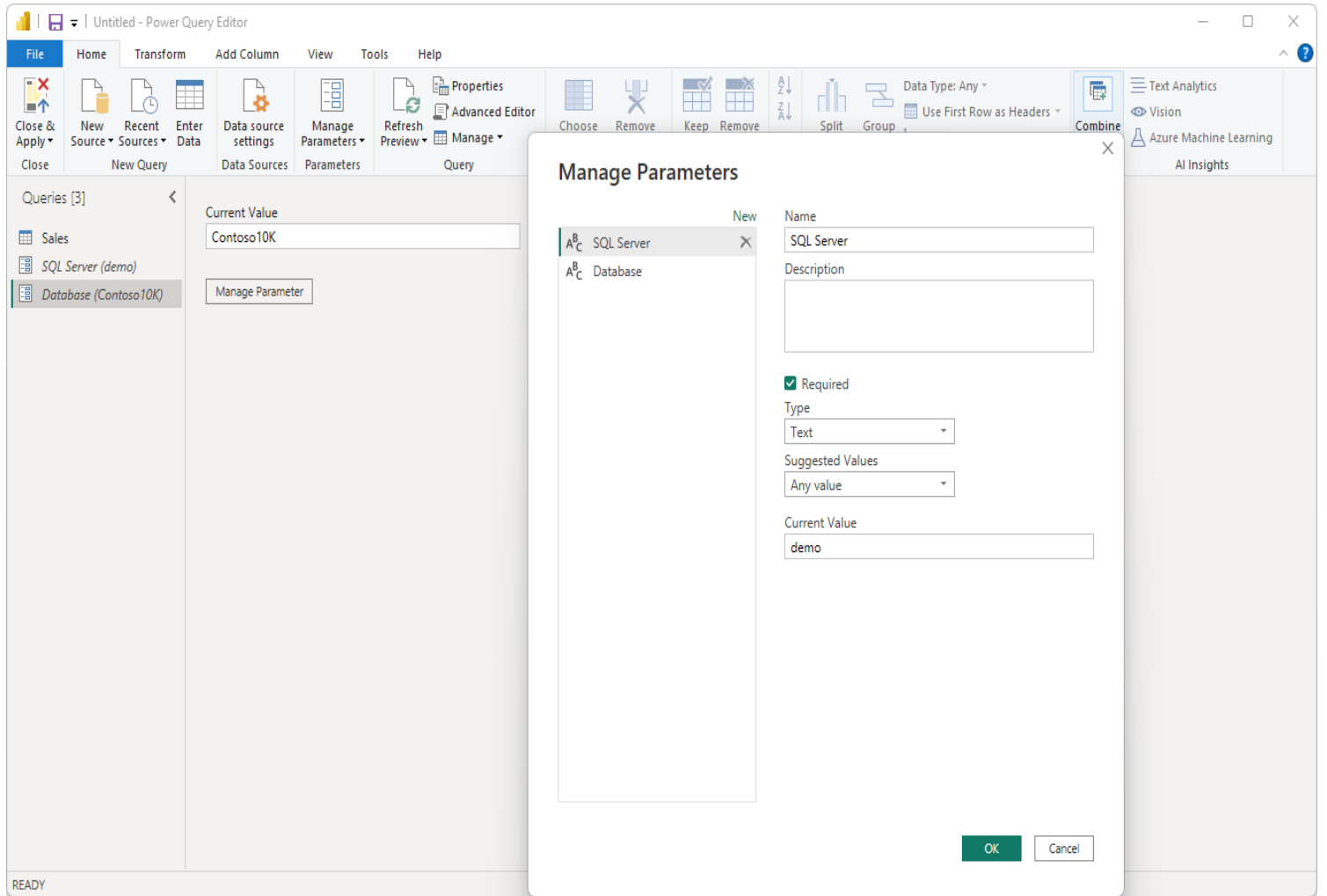


Keep Top 20 Rows does not
SQL Server has a top N clause

Query Folding

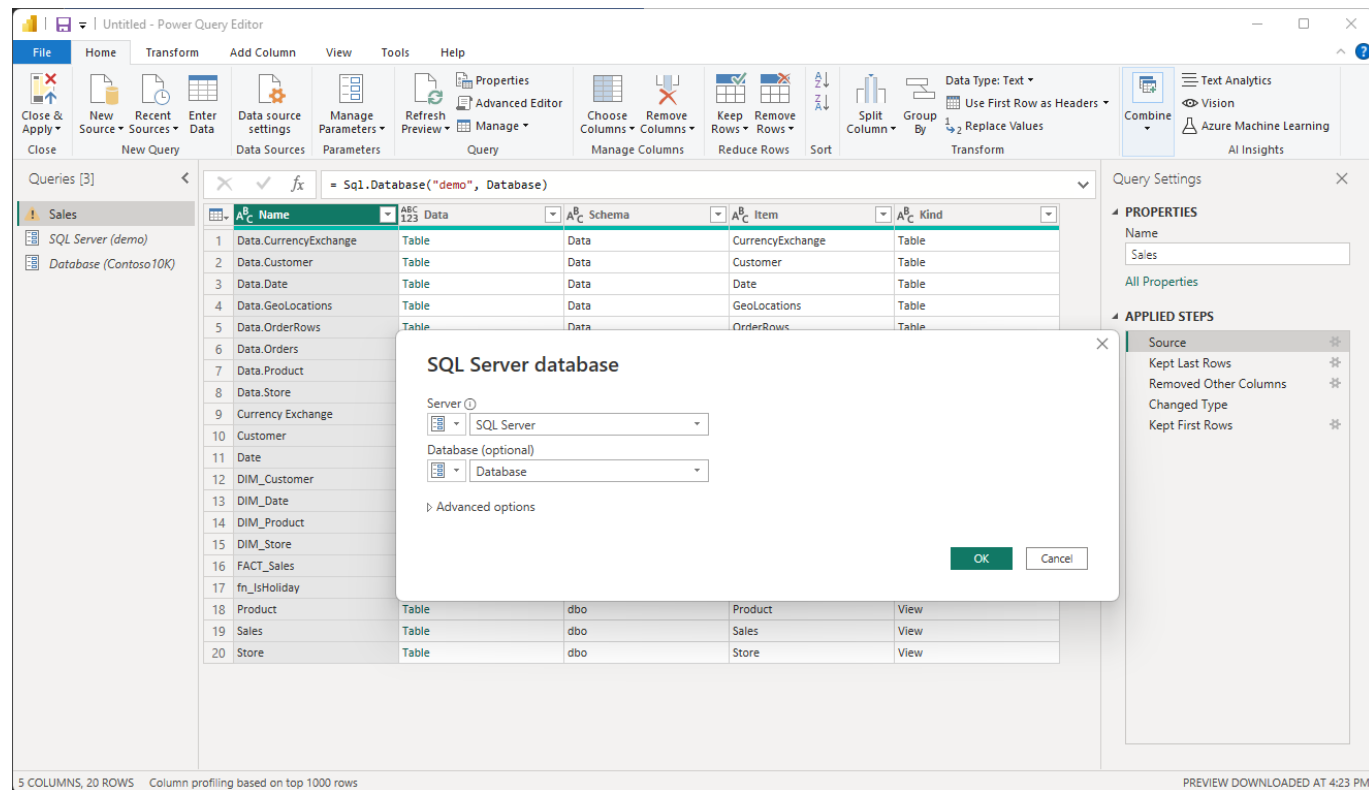
- This is the query sent when the model is refreshed
- This is not the query sent to the tabular engine when you interact with the visuals
- Put off breaking the query folding till the later steps

Parameters



Creating parameters

Using Parameters



Using a parameter for the
Server and Database

Goals of Power Query Steps

- Get smallest dataset possible early in query steps
 - Remove Columns
 - Filter Rows
- Unique Values
- Preserve Query Folding as long as possible

Data Model

Turn off Auto Date Tables

Options

GLOBAL

Data Load

Power Query Editor

DirectQuery

R scripting

Python scripting

Security

Privacy

Regional Settings

Updates

Usage Data

Diagnostics

Preview features

Auto recovery

Report settings

CURRENT FILE

Data Load

Regional Settings

Privacy

Auto recovery

Published dataset settings

Query reduction

Report settings

Background Data

☐ Always allow data previews to download in the background

☒ Allow data previews to download in the background according to each file's setting

☐ Never allow data previews to download in the background

Parallel loading of tables

When you load data into Power BI (via import or DirectQuery), each data table is backed by a Power Query query. These queries are evaluated simultaneously instead of one-by-one, which can speed up the process. In certain situations, you might want to adjust the default number of simultaneous query evaluations and memory used. [Learn more](#)

Maximum number of simultaneous evaluations

20

Maximum memory used per simultaneous evaluation (MB)

432

Time intelligence

☐ Auto date/time for new files [Learn more](#)

Data Cache Management Options

Currently used: 655 KB

Clear Cache

Maximum allowed (MB):

4096

Restore Defaults

Q&A Cache Options

Currently used: 1.43 MB

Clear Cache

Maximum allowed (MB):

4096

OK

Cancel

Hidden Date Tables

- >  Customer
- >  Date
- ✓  *DateTableTemplate_a987aaf9-4324-4fff-9e6b-91eb917a6145*
 -  *Date*
 - >  Date Hierarchy
 - 1²3 Day*
 - A^BC Month*
 - 1²3 MonthNo*
 - A^BC Quarter*
 - 1²3 QuarterNo*
 - 1²3 Year*
- >  *LocalDateTable_329979ec-f948-448d-8d40-da32309c689b*
- >  *LocalDateTable_565d20a7-37da-41eb-acea-3d1681afac62*
- >  *LocalDateTable_6082b466-aa3a-416d-92f5-dee233105c41*
- >  *LocalDateTable_729c8b24-1385-4545-897f-e90fb9602a1d*
- >  *LocalDateTable_86220e95-4654-4888-819a-8afd2ddbe1da*
- >  *LocalDateTable_8ce283be-c847-412c-907b-87781a0d19e1*
- >  Product
- >  Sales
- >  Store

Relationships

Options

GLOBAL

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Auto recovery

Published dataset settings

Query reduction

Report settings

Type Detection

☐ Detect column types and headers for unstructured sources

Relationships

☒ Import relationships from data sources on first load ⓘ

☐ Update or delete relationships when refreshing data ⓘ

☐ Autodetect new relationships after data is loaded ⓘ

[Learn more](#)

Time intelligence

☐ Auto date/time ⓘ [Learn more](#)

Background Data

☒ Allow data previews to download in the background

Parallel loading of tables ⓘ

Maximum number of concurrent jobs [Learn more](#)

☒ Default

☐ One (disable parallel loading)

☐ Custom

Q&A

☐ Turn on Q&A to ask natural language questions about your data ⓘ [Learn more](#)

☐ Share your synonyms with everyone in your org

OK

Cancel

One big wide
table (or tall)

The screenshot shows a table schema in SQL Server Enterprise Manager. The table is named 'SPSubCat_AddCusto...'. The schema lists various columns, including those with aggregate functions (Σ) and those with primary key indicators (PK). The columns are listed in a single column, with a vertical scrollbar on the right side of the list.

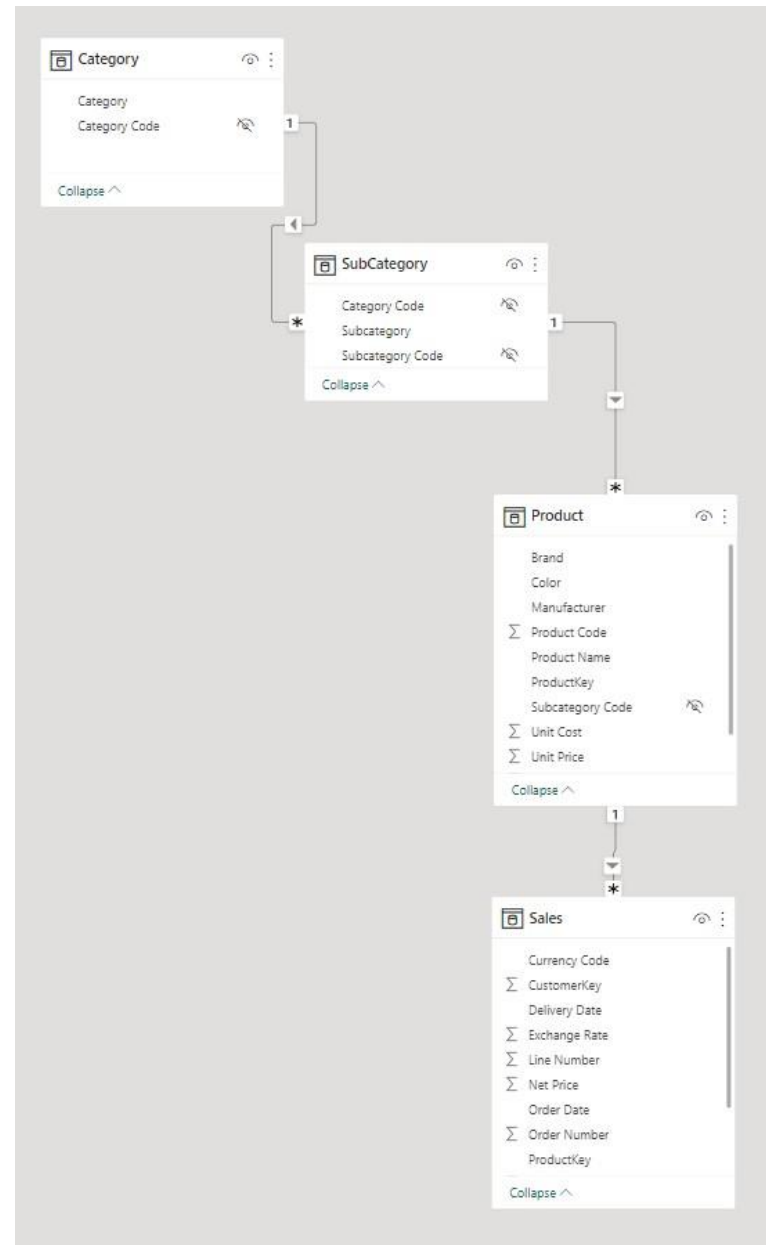
Category.Category
Σ Category.Category Code
Currency Code
Customer.Address
Customer.Age
Customer.Birthday
Customer.City
Customer.Continent
Customer.Country
Customer.Country Code
Customer.CustomerKey
Customer.Gender
Customer.Name
Customer.State
Customer.State Code
Customer.Zip Code
CustomerKey
Delivery Date
Exchange Rate
Line Number
Net Price
Order Date
Order Number
Product.Brand
Σ Product.Category Code
Product.Color
Product.Manufacturer
Product.Product Code
Product.Product Name
Product.ProductKey
Σ Product.Subcategory Code
Product.Unit Cost
Product.Unit Price
Product.Weight
Product.Weight Unit Measure
ProductKey
Quantity
StoreKey

Collapse ^

Wide Table

- Difficult to find the column you are looking for
- Refresh can take longer
- DAX is more complicated
- Performance is better with Star Schema

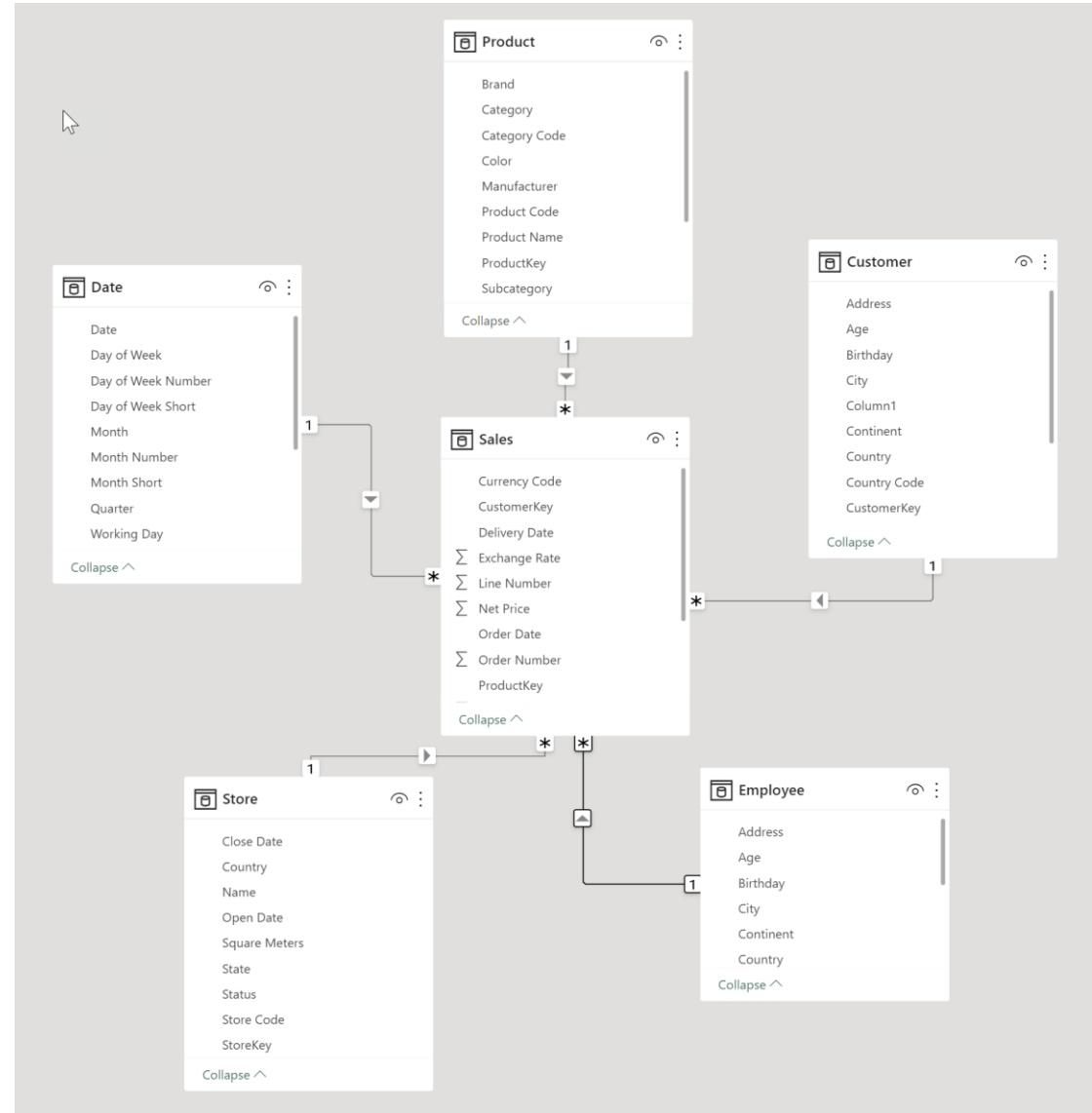
What is Snowflake Schema



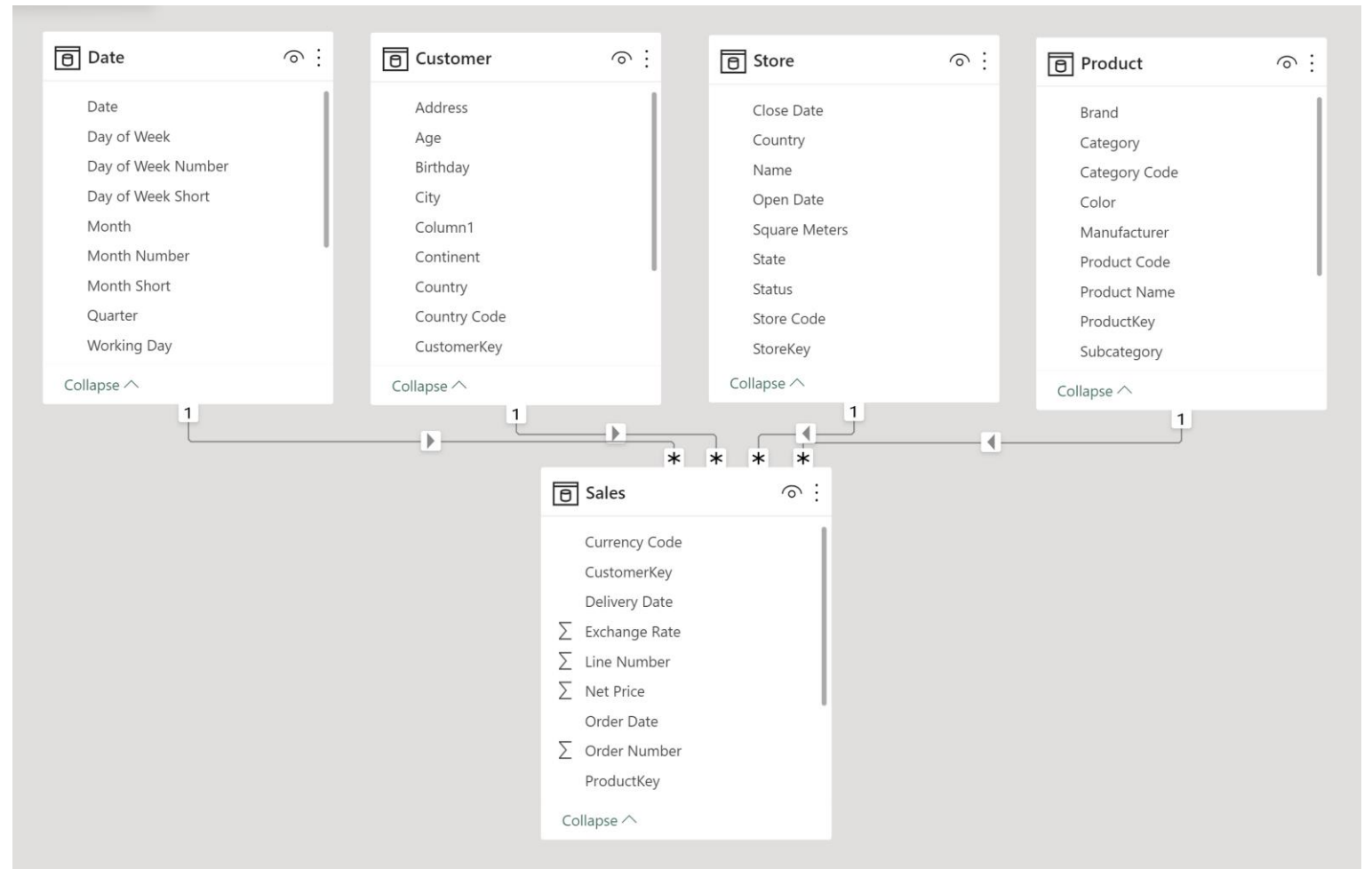
Other examples of Snowflake Schema

- General Ledger
 - Header and Detail
 - Combine into one table
- Employee table with sub type of employee and type of employee.

What is a Star Schema



Collie Method



Snowflake versus Star Schema

Snowflake

- Harder to understand as a developer
- Expanded table issue with DAX

Star Schema

- Simpler
- Overall better performance

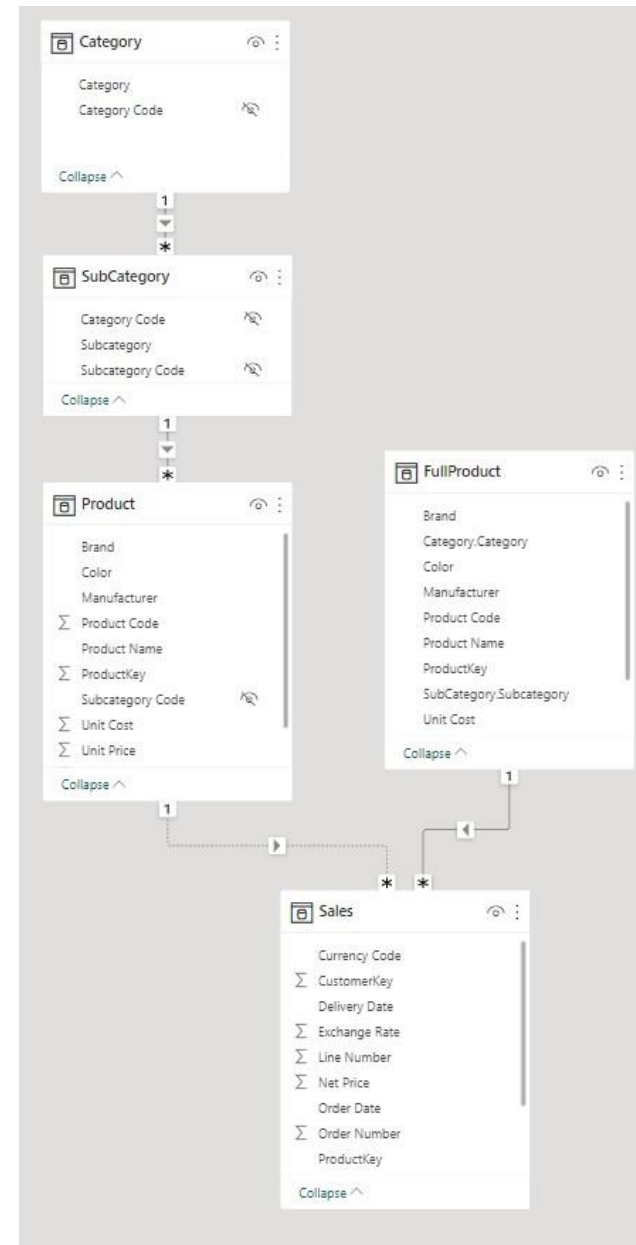
Expanded Tables

- When you apply a table filter it really is the expanded table.

```
CALCULATE (  
    COUNTROWS (Subcategory),  
    Product  
)
```

- It is the expanded table for Product. So only those subcategories that have a product
 - Can't reference base table

Snowflake to (Super) Star Schema



Hiding Columns

- Hide Key Columns
- Build Measures on top of value columns and hide base column
- Implicit Measures versus Explicit Measures

Implicit Measures

- Currently if you create Calculation Groups it disables implicit measures
- Discourage Implicit Measures
 - True/False
- Build explicit Measures

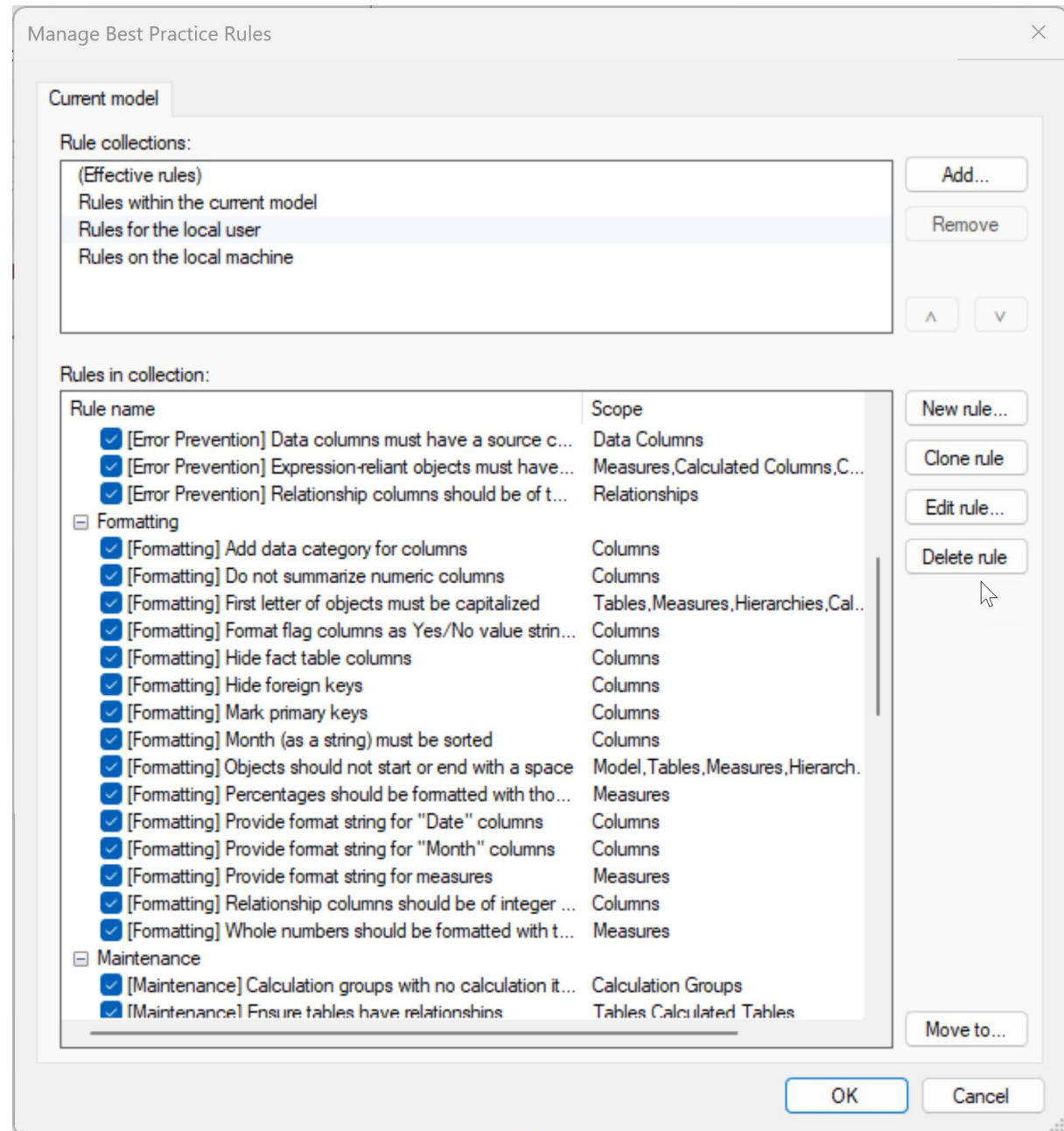
Calculated Columns

- Create all calculated columns in source if possible
- Optimal sorting algorithm only on imported columns

Tabular Editor

- With C# scripts you can automate creating or changing aspects of the model
 - For example, create a measure for every column that has the word amount

Tabular Editor – Best Practice Analyzer



Bravo: Date Table

Bravo does
not support
AutoDate



Unsupported

Models with auto date/time option enabled are not supported.

[Disabling auto date-time in Power BI \(video\)](#)

Time Intelligence

Options

GLOBAL

Data Load

Power Query Editor

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☐ Always allow data previews to download in the background

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☐ Never allow data previews to download in the background

Parallel loading of tables

When you load data into Power BI (via import or DirectQuery), each data table is backed by a Power Query query. These queries are evaluated simultaneously instead of one-by-one, which can speed up the process. In certain situations, you might want to adjust the default number of simultaneous query evaluations and memory used. [Learn more](#)

Maximum number of simultaneous evaluations

Maximum memory used per simultaneous evaluation (MB)

Time intelligence

☐ Auto date/time for new files [Learn more](#)

Data Cache Management Options

Currently used: 655 KB

Clear Cache

Maximum allowed (MB):

Restore Defaults

Q&A Cache Options

Currently used: 1.43 MB

Clear Cache

Maximum allowed (MB):

OK

Cancel

Type of Calendar



Contoso10K > Manage Dates



Model Check



This model **is compatible with the Manage Dates feature.**

You can create new date tables without worrying about breaking measures or reports.

Calendar

Interval

Dates

Holidays

Time Intelligence

Choose a calendar template to apply to this model. Bravo will create the required tables or update them while keeping the existing relationships intact.



Template

Select *Monthly* for calendar based on different number of months. Set *Weekly* for 445-454-544-ISO calendars. Use *Custom* for flexible calendars of variable length. [Manage Templates](#)

Monthly - Fiscal ▾

First Month of the Year

For Weekly ISO, set *January*.

March ▾

Interval

Calendar Interval Dates Holidays Time Intelligence

Select a date interval for your model.



Date Interval

Choose how to determine the date interval of the model. Leave *First Year* and/or *Last Year* empty to use the automatic scan.

First Year

2018

Last Year

2024

Select Measures for Time Intelligence

StarSchema_Date > Manage Dates

Model Check



This model **is compatible with the Manage Dates feature.**

You can create new date tables without worrying about breaking measures or reports.

Calendar Interval Dates Holidays Time Intelligence

Create the most common Time Intelligence DAX functions available in your model.



Time Intelligence Functions



Target Measures

Choose the measure used to generate the Time Intelligence functions.

Choose Measures... ▾



Search



TABLE \ COLUMN



Sales

Time Intelligence Measures

<  StarSchema_Date > Manage Dates > Preview

Dates Time Intelligence



▼  To-date growth	A
▼  Sales Amount	A
 MOMTD % Sales Amount	A
 MOMTD Sales Amount	A
 PMTD Sales Amount	A
 PQTD Sales Amount	A
 PYTD Sales Amount	A
 QOQTD % Sales Amount	A
 QOQTD Sales Amount	A
 YOYTD % Sales Amount	A
 YOYTD Sales Amount	A
▼  To-date total	A
▼  Sales Amount	A
 MTD Sales Amount	A
 QTD Sales Amount	A
 YTD Sales Amount	A

Bravo: Optimize Model

Find unused columns, size of columns, and unique elements

Bravo Optimize Model

StarSchema

Your dataset is **1.48 MB** large and contains **66** columns, **56** of which are not referenced within the model.

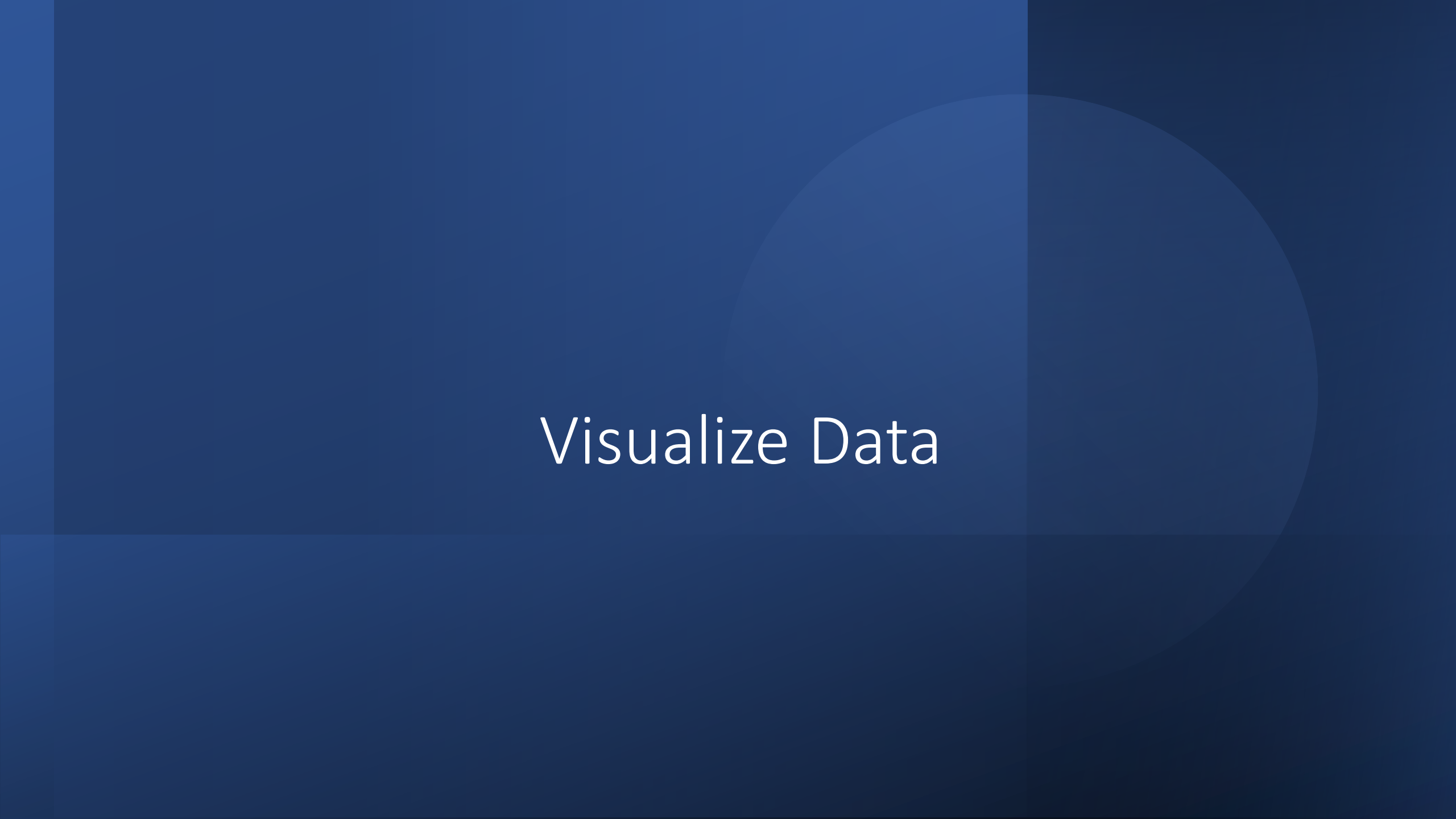
Search Column

VPAX

	COLUMN	TABLE	CARDINALITY	SIZE	WEIG...
	Smaller columns...		10.44 K	1.08 MB	73%
	Product Name	Product	2.52 K	145.70 KB	10%
123	ProductKey	Product	2.52 K	97.86 KB	6%
	Product Code	Product	2.52 K	84.35 KB	6%
	Date	Date	1.10 K	41.05 KB	3%
	Birthday	Customer	511	40.34 KB	3%
			19.60 K	1.48 MB	100%

🔗 Unreferenced columns can generally be removed from the model to optimize performance. Before removing them, make sure you are not using these columns in any reports, which Bravo cannot determine.



The background is a solid dark blue. A large, semi-transparent circle of a slightly lighter blue shade is positioned on the right side, partially overlapping a vertical line of the same lighter blue color that runs from the top to the bottom of the frame.

Visualize Data

Conclusion

- Shape data as much as possible in Power Query
- Import it into a Star Schema
- Use external tools to help optimize model

Resources

- [Guide to Query Folding](#)
- [Improve Model Performance](#)
- [Bravo for Power BI](#)
- [Tabular Editor 2](#)

SQLOrlando



SQLSATURDAY

OCT, 7th 2023

Orlando, FL

Scan the QR code on
the room poster to
fill out session and
event evaluations



Thank you

Jason Romans

thedaxshepherd@gmail.com

<https://thedaxshepherd.net/> 

<https://www.linkedin.com/in/jason-r-sql-jar>